TEACH YOURSELF INTERMEDIATE XI UNIT 4

# Data and Analysis (Databases)

Relational design, integrity, and DBMS objects

### What you will master

- Define tables, keys, and relationships (1:1, 1:M, M:N)
- Apply referential integrity (cascade update/delete)
- Draw ER diagrams and relational schemas
- Use tables, forms, queries, and reports in a DBMS

### How to use these notes

- Read one section, then close the book and explain it aloud.
- Do every *Try it* box before moving on.
- Memorize tables and truth patterns — exams love them.

## 4.1 Database concepts

A **database** stores related data persistently. A **DBMS** (e.g. Access, MySQL) manages storage, queries, and security.

**Primary key** uniquely identifies a row. **Foreign key** links to another table's primary key.




Figure 4.1 — Students enroll in many courses (M:N via junction table).

## 4.2 Relationships & integrity

| Type | Meaning | Example |
|---|---|---|
| 1:1 | One row ↔ one row | Person ↔ Passport |
| 1:M | One parent, many children | Department ↔ Employees |
| M:N | Many on both sides | Students ↔ Courses |

**Referential integrity** prevents orphan records. **Cascade update/delete** propagates changes when parent rows change.

## 4.3 Queries & reports

**Queries** filter and calculate (SELECT, WHERE, ORDER BY). **Forms** ease data entry; **reports** present summaries for printing.

**Try it**

Design tables Student(ID, Name) and Result(StudentID, Subject, Marks) with a 1:M link. Write a query for students with Marks > 80.

## Teach Yourself — Practice & Check

1. Convert an M:N Teacher–Subject assignment into two 1:M tables.
2. List fields you would put in an ER diagram for a library system.